

Tutorial 4

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Subject : OOP

Tutorial Title : Write a c++ program to perform addition, subtraction, multiplication and division on 2 complex numbers using the concept of class.

Program:

```
#include<iostream>
```

```
using namespace std;
```

```
class complex {  
    float real, img;
```

```
public:
```

```
    void accept(float r, float i) {  
        real = r;  
        img = i;  
    }
```

```
    void display() {  
        if (img >= 0) {  
            cout << real << " + " << img << "i";  
        } else {  
            cout << real << " - " << -img << "i";  
        }  
    }
```

```
    void add(complex c1, complex c2) {  
        real = c1.real + c2.real;  
        img = c1.img + c2.img;  
    }
```

```
    void sub(complex c1, complex c2) {  
        real = c1.real - c2.real;  
        img = c1.img - c2.img;  
    }
```

```
    void mult(complex c1, complex c2) {  
  
        real = (c1.real * c2.real) - (c1.img * c2.img);
```

```

        img = (c1.real * c2.img) + (c1.img * c2.real);
    }

    void divide(complex c1, complex c2) {
        float denominator = (c2.real * c2.real) + (c2.img * c2.img);
        real = (c1.real * c2.real + c1.img * c2.img) / denominator;
        img = (c1.img * c2.real - c1.real * c2.img) / denominator;
    }
};

int main() {
    complex c1, c2, c3, c4, c5, c6;

    c1.accept(3, -5);
    c2.accept(9, 20);

    cout << "c1 = ";
    c1.display();
    cout << "\nc2 = ";
    c2.display();

    c3.add(c1, c2);
    cout << "\n\nAddition: ";
    c3.display();

    c4.sub(c1, c2);
    cout << "\nSubtraction: ";
    c4.display();

    c5.mult(c1, c2);
    cout << "\nMultiplication: ";
    c5.display();

    c6.divide(c1, c2);
    cout << "\nDivision: ";
    c6.display();

    cout << endl;
    return 0;
}

```

Output:

$$c1 = 3 - 5i$$

$$c2 = 9 + 20i$$

$$\text{Addition: } 12 + 15i$$

$$\text{Subtraction: } -6 - 25i$$

$$\text{Multiplication: } 127 + 15i$$

$$\text{Division: } -0.151767 - 0.218295i$$